

Project Information

Project: Census Income Prediction with CatBoost

Dataset: UCI Adult (Census Income) Dataset

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Model: CatBoostClassifier

1. Executive Summary

This project develops a CatBoost-based classification model to predict whether an individual's annual income exceeds \$50,000 using demographic, educational, and occupational attributes from the UCI Adult dataset.

The dataset contains 48,842 records and 14 predictor variables, including age, education, occupation, marital status, and work-related information. CatBoost was selected because it handles categorical features natively without requiring extensive one-hot encoding.

The final tuned model achieved:

- ROC-AUC: 0.9275
- Precision: 70.86%
- Recall: 73.92%
- F1-Score: 72.36%
- Accuracy: 86.66%

Threshold optimization identified **0.40** as the best classification threshold, providing the highest F1-score and improved recall compared to the default threshold.

Key findings indicate that capital gain, relationship status, and age are among the strongest predictors of income. While the model performs consistently across demographic groups, performance differences exist and should be monitored. This project is intended for educational purposes only and should not be used for real-world hiring, lending, or other high-impact decisions.

2. Background & Problem Definition

The Adult Census Income dataset is commonly used to study income prediction using machine learning techniques. The objective is to classify individuals into one of two categories:

- Income > \$50K
- Income ≤ \$50K

A False Positive occurs when the model predicts income > \$50K for an individual whose actual income is ≤ \$50K. A False Negative occurs when the model predicts income ≤ \$50K for an individual whose actual income exceeds \$50K.

Because classification decisions depend on probability thresholds, threshold selection plays an important role in balancing precision and recall. Therefore, threshold optimization was performed to identify the most suitable operating point.

3. Dataset Summary & Data Quality Checks

The UCI Adult dataset contains **48,842 records**, **14 predictor variables**, and one target variable representing income category.

The dataset includes both numerical features (age, hours_per_week, capital_gain) and categorical features (workclass, education, occupation, marital_status, relationship, race, sex, native_country).

Missing values were identified in:

- workclass
- occupation
- native_country

These missing values are represented using the symbol "?" and were retained for CatBoost's native categorical processing. Duplicate records were removed during preprocessing.

Column	Type	Missing Handling
Numerical Features	Integer	No missing values
Categorical Features	Category	Retained for CatBoost
Income	Target	Binary encoded (0/1)

4. Preprocessing Policy

Data preprocessing was performed to ensure data quality and reproducibility before model training. Missing values represented by the symbol "?" were retained as categorical values and handled using CatBoost's native categorical processing capabilities. Duplicate records identified in the training dataset were removed to avoid overrepresentation of repeated observations.

Categorical variables were passed directly to CatBoost without applying one-hot encoding. This approach preserves the original feature information while leveraging CatBoost's built-in handling of categorical data.

For reproducibility, the dataset was divided into training, validation, and testing sets using a fixed random seed of **42**. The validation set was used for hyperparameter tuning and early stopping, while the test set remained untouched until final evaluation.

5. Model Development (CatBoostClassifier)

CatBoostClassifier was selected as the sole supervised learning model because it performs effectively on tabular datasets containing both numerical and categorical features.

The baseline model was trained using:

- Loss Function: Logloss
- Evaluation Metric: AUC

- Random Seed: 42
- Early Stopping: Enabled using the validation dataset

Categorical features were specified using CatBoost's **cat_features** parameter, allowing the model to process categorical variables directly without additional encoding. This reduced preprocessing complexity and improved the efficiency of model training.

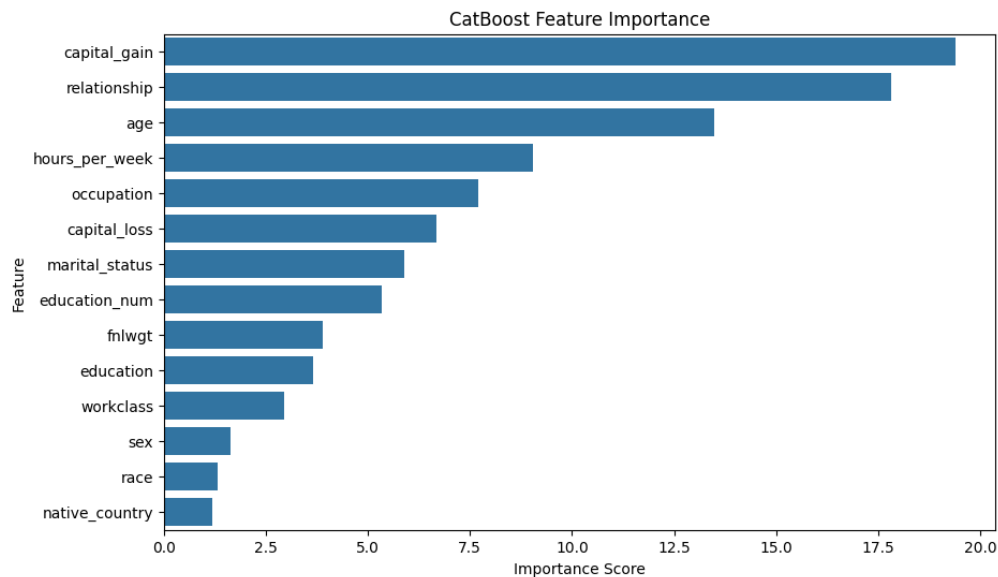


Figure 1. CatBoost Feature Importance Ranking

6. Hyperparameter Tuning

Hyperparameter tuning was performed to improve the baseline CatBoost model and enhance generalization performance. A **Randomized Search** approach was selected instead of exhaustive Grid Search because it explores a wide range of parameter combinations while significantly reducing computational cost.

Model configurations were evaluated using the validation dataset. **Validation AUC** was used as the primary selection metric because it measures the model's ability to distinguish between income classes independent of a specific classification threshold. **F1-Score** was considered as a secondary metric to ensure a balanced trade-off between precision and recall.

Final Hyperparameters

Parameter	Final Value
iterations	1000
learning_rate	0.05
depth	6
l2_leaf_reg	5
bagging_temperature	0

random_strength	1
one_hot_max_size	2

The selected configuration achieved the highest validation AUC while maintaining strong classification performance. Early stopping was enabled during training to prevent overfitting and automatically determine the optimal number of boosting iterations.

7. Evaluation (Final Test Set)

7.1 Performance Metrics

The final tuned CatBoost model was evaluated on the untouched test dataset using the optimized classification threshold of 0.40.

Metric	Value
ROC-AUC	0.9275
Accuracy	86.66%
Precision	70.86%
Recall	73.92%
F1-Score	72.36%

The results indicate strong classification performance and excellent class separability. The high ROC-AUC demonstrates the model's ability to distinguish between high-income and low-income individuals, while the balanced precision and recall values indicate effective identification of the positive class.

7.2 Confusion Matrix

The confusion matrix provides a detailed breakdown of classification outcomes.

- True Positives (TP): Correctly predicted income > \$50K
- True Negatives (TN): Correctly predicted income ≤ \$50K
- False Positives (FP): Predicted income > \$50K when actual income ≤ \$50K
- False Negatives (FN): Predicted income ≤ \$50K when actual income > \$50K

False Positives may lead to overestimating an individual's income category, while False Negatives represent missed high-income individuals.

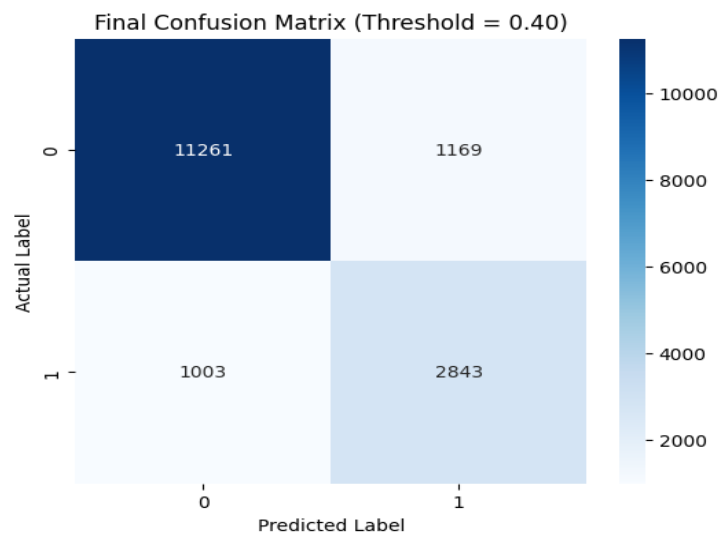


Figure 2. Confusion Matrix on the Test Dataset

7.3 ROC Curve

The Receiver Operating Characteristic (ROC) Curve evaluates model performance across all possible classification thresholds.

The model achieved a ROC-AUC score of **0.9275**, indicating excellent discrimination capability. This suggests that the model can correctly rank high-income individuals above low-income individuals approximately 92.75% of the time.

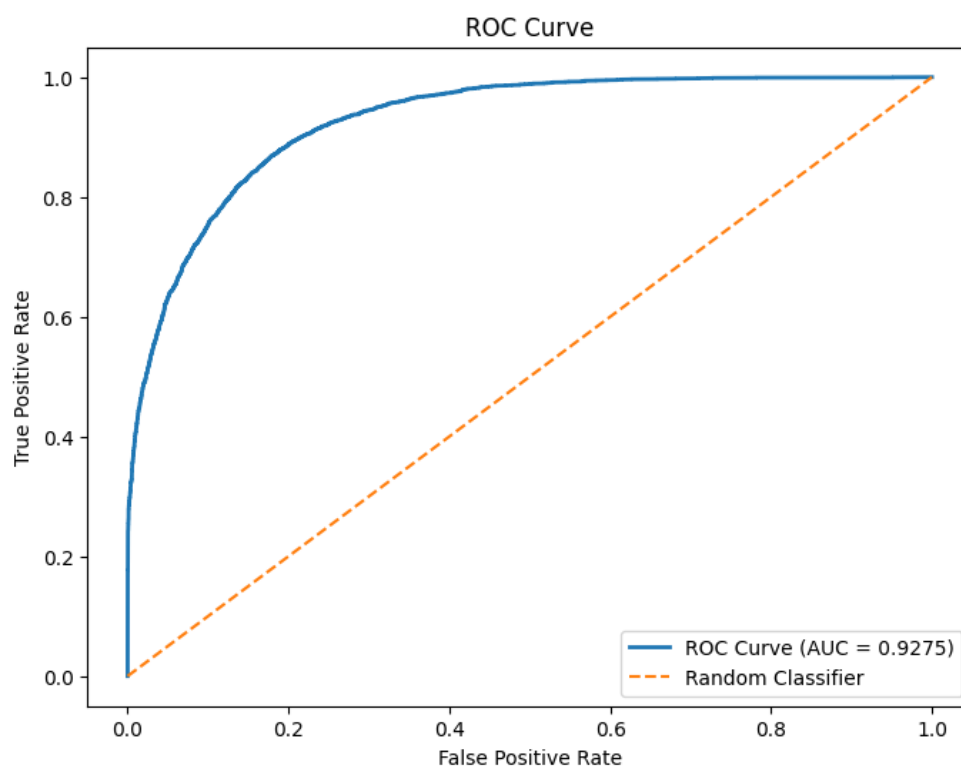


Figure 3. ROC Curve for the Tuned CatBoost Model

7.4 Threshold Selection

Multiple classification thresholds were evaluated to balance precision and recall. The default threshold of 0.50 produced higher precision but lower recall.

A threshold of **0.40** was selected because it achieved the highest F1-Score and improved the identification of high-income individuals while maintaining acceptable precision.

This threshold provides a better balance between False Positives and False Negatives and was therefore used for final model evaluation.

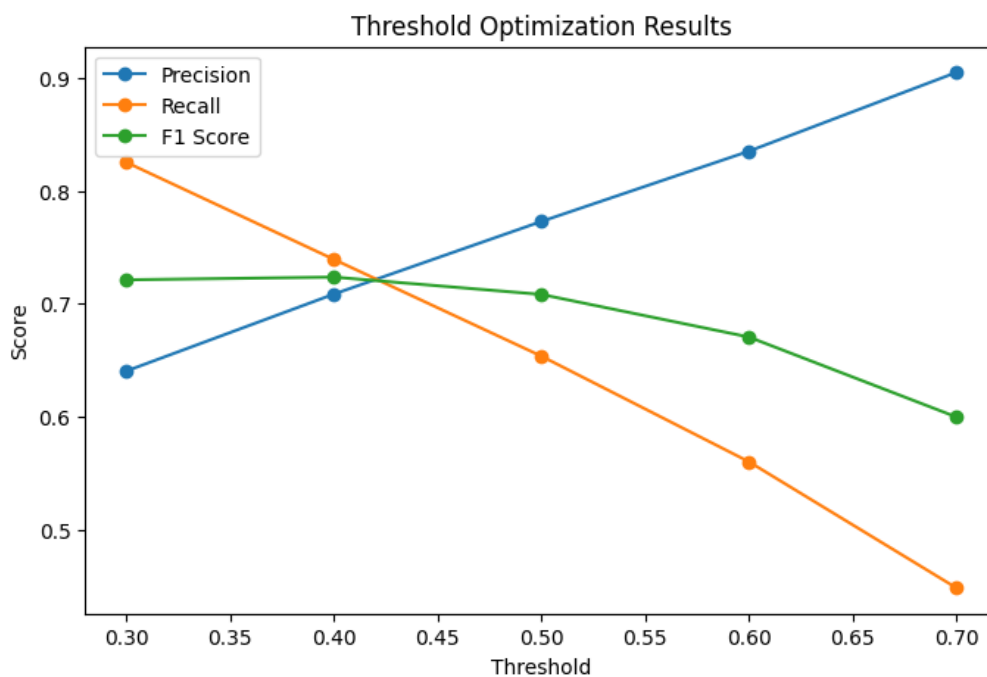


Figure 4. Precision, Recall, and F1-Score Across Classification Thresholds

8. Error Analysis

Error analysis was conducted to understand where the model makes incorrect predictions and identify opportunities for improvement.

False Positive Patterns

False Positives were frequently observed among individuals with:

- Professional and managerial occupations
- Bachelor's level education
- Married-civ-spouse marital status
- Moderate capital gain values

These individuals exhibit characteristics commonly associated with higher income, causing the model to occasionally overestimate income class.

False Negative Patterns

False Negatives were commonly observed among individuals with:

- Skilled trade occupations such as Craft-repair
- Bachelor's degree holders
- Married-civ-spouse individuals
- Limited capital gain despite relatively strong demographic indicators

These cases suggest that some high-income individuals do not follow the typical income patterns learned by the model.

Key Insights

1. Capital gain is the strongest driver of income prediction.
2. Education alone is insufficient to accurately determine income level.
3. Occupation and relationship status strongly influence classification decisions.
4. Some income groups exhibit overlapping characteristics, making classification challenging.

Potential Improvements

1. Incorporate additional socioeconomic variables if available.
2. Explore advanced threshold optimization strategies for specific business objectives.
3. Investigate calibration techniques to improve probability estimates and reduce classification errors.

9. Responsible ML Checks

Because the Adult dataset contains sensitive demographic attributes such as sex and race, group-wise performance analysis was conducted to evaluate whether model behavior differs across demographic groups.

Performance by Sex

Sex	AUC	Recall	FPR	FNR
Female	0.9460	0.6627	0.0315	0.3373
Male	0.9098	0.7531	0.1338	0.2469

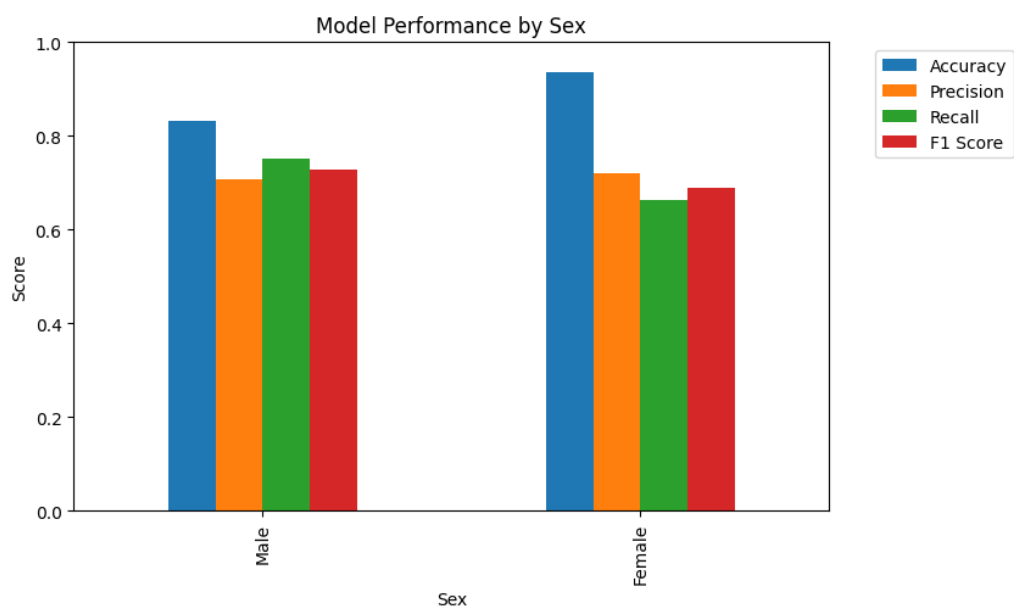


Figure 5. Group-wise Performance Metrics by Sex

Performance by Race

Race	AUC	Recall	FPR	FNR
White	0.9245	0.7278	0.1031	0.2722
Black	0.9453	0.7303	0.0685	0.2697
Asian-Pac-Islander	0.9516	0.7519	0.1182	0.2481
Amer-Indian-Eskimo	0.9189	0.7368	0.0702	0.2632
Other	0.9067	0.6000	0.0568	0.4000

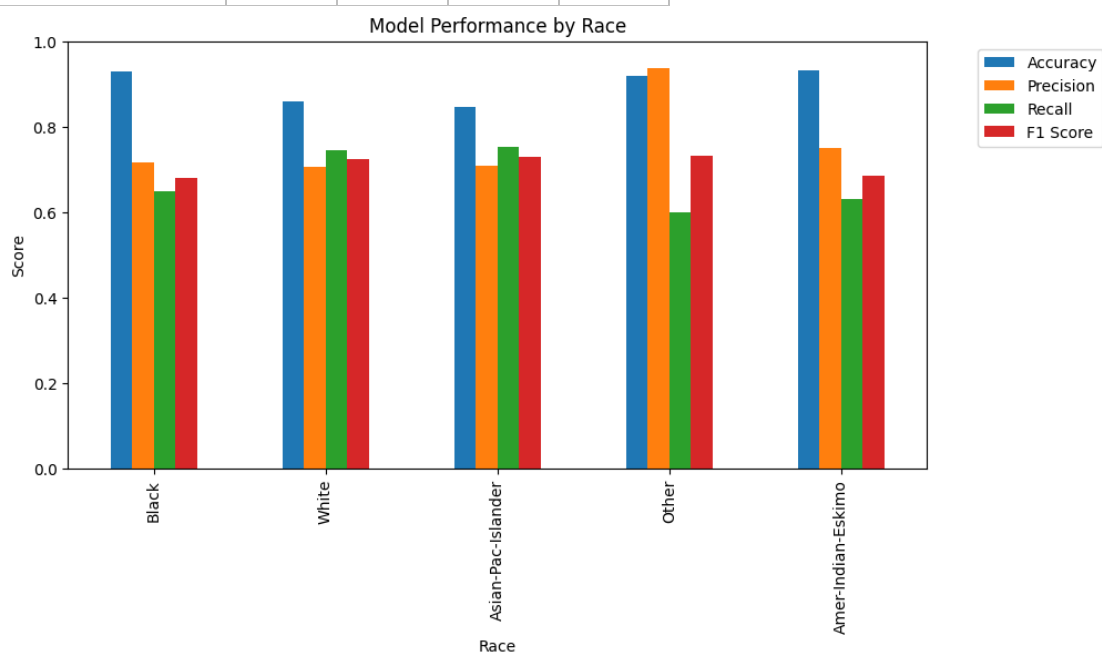


Figure 6. Group-wise Performance Metrics by Race

Discussion

Performance differences across demographic groups may arise due to differences in sample size, class distributions, feature distributions, and historical socioeconomic patterns present in the dataset. Although the model demonstrates strong predictive performance overall, variations in recall and error rates indicate that model behavior is not identical across all groups.

Therefore, this model should be viewed as an educational exercise and should not be used for real-world employment, lending, admissions, or other high-impact decision-making scenarios without appropriate governance, fairness auditing, monitoring, and regulatory oversight.

10. Recommendations, Limitations, and Next Steps

Recommendations

1. Continue using CatBoost for tabular datasets containing mixed numerical and categorical features.
2. Monitor Recall and F1-Score in addition to Accuracy when evaluating future model versions.
3. Retain threshold optimization as part of the model development workflow.
4. Investigate False Negative cases to improve identification of high-income individuals.
5. Perform periodic fairness assessments across demographic groups.
6. Use probability-based predictions rather than relying solely on class labels.
7. Consider additional socioeconomic variables if available.
8. Maintain reproducible preprocessing and model training procedures.

Limitations

- The Adult dataset represents historical census information and may not reflect current economic conditions.
- Missing values exist in several categorical attributes and may influence model behavior.
- Certain variables may act as proxy indicators for sensitive demographic characteristics.
- Performance differences across demographic groups highlight potential fairness concerns.
- The dataset does not contain all factors that may influence income.

Next Steps

- Collect more recent and representative data.
- Perform deeper fairness auditing and bias analysis.
- Implement ongoing model monitoring and performance tracking.
- Develop model documentation and model cards for transparency.
- Explore calibration techniques and advanced error reduction strategies.